Appendix B: Backend route excerpt

app.py

Verified 15 June 2026: Python compileall and dashboard Node.js syntax check passed. Lines beginning '| ' are visual wraps.

```
import os
import threading
import webbrowser
from datetime import datetime
from pathlib import Path

from flask import Flask, jsonify, render_template, request
from data_source import (
    ICAO_EXCEL_DIR,
    SESSION_DIR,
    SIMULATION_MODE,
    get_aircraft_data,
    get_status,
    set_simulation_mode,
)
from preprocessing.pingstation_preprocess import PingstationPreprocessor
from storage.flight_data_store import FlightTrajectoryStore
from runtime_paths import app_dir, resource_dir
from aircraft_lookup import lookup_aircraft_info, lookup_flight_route

app = Flask(
    __name__,
    template_folder=str(resource_dir() / "templates"),
    static_folder=str(resource_dir() / "static"),
)
preprocessor = PingstationPreprocessor()
flight_store = FlightTrajectoryStore(
    db_path=os.getenv(
        "FLIGHT_DB_PATH",
        str(app_dir() / "logs" / "flight_training.db"),
    ),
    enabled=os.getenv("FLIGHT_DB_ENABLED", "true").lower() == "true",
    min_interval_sec=int(os.getenv("FLIGHT_DB_MIN_INTERVAL_SEC", "5")),
    min_distance_m=float(os.getenv("FLIGHT_DB_MIN_DISTANCE_M", "30")),
    min_altitude_change_m=int(os.getenv("FLIGHT_DB_MIN_ALT_CHANGE_M", "20")),
    min_heading_change_de2=int(os.getenv("FLIGHT_DB_MIN_HEADING_CHANGE_DE2", "500")),
    min_speed_change_cms=int(os.getenv("FLIGHT_DB_MIN_SPEED_CHANGE_CMS", "200")),
    history_retention_days=int(os.getenv("FLIGHT_DB_HISTORY_RETENTION_DAYS", "30")),
    append_csv_enabled=os.getenv("FLIGHT_TRAJECTORY_APPEND_CSV_ENABLED", "true").lower() == "true",
    append_csv_dir=os.getenv(
        "FLIGHT_TRAJECTORY_APPEND_CSV_DIR",
        str(app_dir() / "logs" / "training_csv"),
    ),
    icao_excel_enabled=os.getenv("FLIGHT_ICAO_EXCEL_ENABLED", "true").lower() == "true",
    icao_excel_dir=os.getenv(
        "FLIGHT_ICAO_EXCEL_DIR",
        str(ICAO_EXCEL_DIR),
    ),
    icao_master_enabled=os.getenv("FLIGHT_ICAO_MASTER_ENABLED", "true").lower() == "true",
    icao_master_path=os.getenv(
        "FLIGHT_ICAO_MASTER_PATH",
        str(SESSION_DIR / "MASTER_ICAO.xlsx"),
    ),
    reset_excel_on_start=os.getenv("FLIGHT_ICAO_RESET_ON_START", "false").lower() == "true",
    icao_excel_extension=os.getenv("FLIGHT_ICAO_EXCEL_EXTENSION", "xls"),
)
matrix_csv_enabled = os.getenv("FLIGHT_MATRIX_CSV_ENABLED", "false").lower() == "true"
matrix_csv_path = os.getenv(
    "FLIGHT_MATRIX_CSV_PATH",
    str(app_dir() / "logs" / "flight_matrix_latest.csv"),
)
matrix_csv_limit = int(os.getenv("FLIGHT_MATRIX_CSV_LIMIT", "5000"))
prediction_enabled = os.getenv("FLIGHT_PREDICTION_ENABLED", "true").lower() == "true"
prediction_max_age_sec = int(os.getenv("FLIGHT_PREDICTION_MAX_AGE_SEC", "180"))
prediction_limit = int(os.getenv("FLIGHT_PREDICTION_LIMIT", "1000"))

if SIMULATION_MODE:
    # Simulation timestamps are synthetic and restart from the same base each
```